INTERNATIONAL MONETARY FUND

SENEGAL

November 15, 2016

STAFF REPORT FOR THE 2016 ARTICLE IV CONSULTATION AND THIRD REVIEW UNDER THE POLICY SUPPORT INSTRUMENT—DEBT SUSTAINABILITY ANALYSIS

Approved By
Roger Nord and Peter Allum (IMF), and Paloma Anos-Casero (IDA)

Prepared by the staffs of the International Monetary Fund and the International Development Association

Risk of external debt distress	Low
Augmented by significant risks stemming from domestic public and/or private external debt?	No

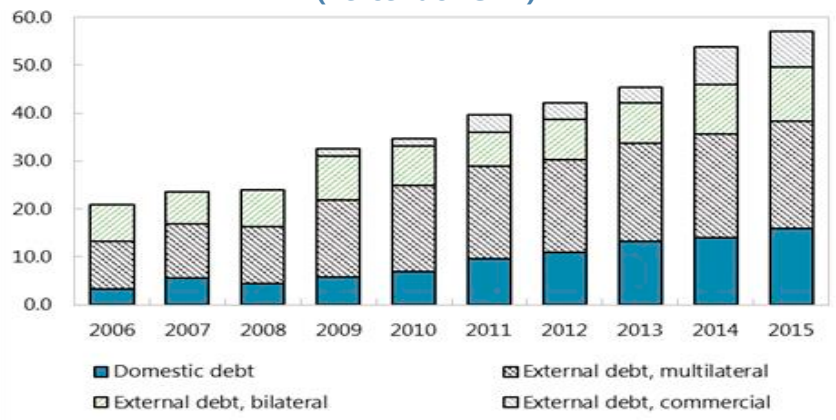
Senegal remains at low risk of debt distress, consistent with the Staff Report of January 2016 (EBS/16/03).[1] All debt burden indicators are below their respective thresholds, with only the debt service-to-revenue ratio showing two spikes that breach the threshold under stress scenarios due to Eurobond rollover. These breaches are small and temporary reflecting liquidity rather than solvency issues. However, public debt and debt service have been rising in recent years. The low risk of debt distress is predicated on adherence to the planned fiscal consolidation path, an acceleration of reforms under the Plan Sénégal Emergent (PSE), and limited non-concessional borrowing. The public DSA does not find significant weaknesses, but the persistent high debt service as a share of revenue calls for strengthening debt management and prudent borrowing strategies.

[1] Senegal's public debt statistics cover external and domestic debt issued by the central government (including debt guaranteed by the government). External debt is defined as debt borrowed or serviced in a currency other than the CFA franc, regardless of the residency of the creditor. The baseline DSA incorporates remittances, as they represent a significant share of GDP (12.1 percent over 2013-2015) and exports of goods (61 percent over 2013-2015). Senegal's policy performance is ranked "strong" by the CPIA with a score of 3.8.

BACKGROUND

1. Public debt ratios have been revised upwards in 2016, but they remain on a declining path over the medium term. Historical figures on the composition of the stock of external debt have not been revised with respect to the ones published in the last DSA update (EBS/16/03). For 2016, external public debt in Senegal is projected at 39.9 percent of GDP, compared to 39.3 percent projected in the previous DSA. The reasons for this difference are changes in nominal GDP and a small increase in external borrowing. Commercial debt stood at 20 percent of the stock of total external public debt in 2014 and declined to 17.8 percent in June 2016. Domestic debt is projected to increase from 15.8 percent of GDP (CFAF 1,280 billion) at end-2015 to 19.2 percent of GDP (CFAF 1,681 billion) at end-2016, reflecting the accumulation of additional domestic debt, related to Treasury operations (see also Staff Report). This projection of domestic debt for 2016 is significantly higher than 15.8 percent projected in the latest DSA. As a result, total public debt is projected to reach 59.1 percent of GDP (CFAF 5,152 billion), 4 percent of GDP more than what was estimated in the previous DSA (EBS/16/03). Moreover, debt service on total public debt reached 25.4 percent of revenue in 2016 and is projected to increase significantly to 36 percent of revenue in 2018.

Text Figure 1. Senegal: Public Debt, 2006-15
(Percent of GDP)



2. The authorities are committed to reduce debt ratios over the medium term, thanks to a continuous effort of fiscal consolidation, improvements in the current account and a strengthening of debt management policies—including more attention to the terms and volume of non-concessional financing. The authorities are increasingly financing government needs on the internal and regional markets. On the one hand, this strategy is welcome given that it mitigates exchange rate risks and the vulnerability to the volatility of external capital flows. In addition, Senegal has been able to lengthen the maturities and reduce borrowing costs on domestic debt. The average maturity on domestic debt increased from 22 months in 2011 to 57 months in 2015, thanks to issuances of 10-year bonds in 2013 and 2015. Interest rates on one-year Treasury Bills declined from 6.2 percent in 2011 to 4.9 in 2015, while rates on five-year T-Bonds moved from 9.1 to 5.9 percent between 2012 and 2015. On the other

hand, domestic borrowing is still more expensive than semi-concessional and concessional lending and some recent loans carry interest rates as high as 8.5 percent, undermining future debt sustainability. As a result, the authorities are committed to a debt management strategy that will rely predominantly on concessional and semi-concessional borrowing, largely from traditional bilateral and multilateral lenders, and resort to non-concessional borrowing only in exceptional cases and for specific high-return projects. Eurobond issuances and, more generally, borrowing on commercial terms, will be considered if financing terms are favorable and if it is not possible to obtain non-concessional financing from development partners, particularly the African Development Bank and the World Bank.

UNDERLYING ASSUMPTIONS AND BORROWING PLAN

3. The DSA is consistent with the macroeconomic framework outlined in the Staff Report and updates the previous DSA produced in EBS/16/03, for the first review of the three-year Policy Support Instrument (PSI). In line with the previous DSA, the baseline scenario assumes the implementation of sound macroeconomic policies, structural reforms, and an ambitious investment plan, as outlined in the PSE. This scenario is expected to deliver strong and sustained economic growth and a narrowing fiscal deficit over the medium term. As a result, medium-term projections are significantly better than historical averages—especially for real GDP growth and the primary balance—but aligned with recent outcomes in 2014-15, implying that the take-off is assumed to persist, conditional on reforms being implemented. The main assumptions are as follows:

- **Real GDP growth** is estimated at 6.6 percent in 2016 and will increase to 7.1 percent in 2021 to reflect the effects of infrastructure investment, export growth, and reforms under the PSE. Over the long run, real GDP growth is projected at 5.5 percent over the period 2022-36, slightly lower than in the previous DSA. This is in line with the international experience that suggests that over a long period, as economies converge to middle income status, the real growth rate slows down.

- **Fiscal deficit**. The overall fiscal deficit is projected at 4.2 percent of GDP in 2016, 3.7 percent in 2017, and 3.0 percent in 2018. In the long run, the deficit is set at 3.0 percent of GDP, revised slightly upwards compared to the previous DSA to reflect the presence of investment needs to meet development challenges.

- **Current account deficit**. The current account deficit is projected to continue its recent downward trend in 2016, reaching 6.5 percent of GDP. Starting in 2017, the current account is projected to slightly worsen because of increasing FDI-related imports, but in the medium term higher FDI-related exports will drive an improvement in the external position. Over the long-term, the average current account deficit should stabilize at 6.7 percent of GDP, about one percentage point less than what was expected in the previous DSA, mostly because of more favorable terms of trade. Remittances remain a significant component of the current account—estimated at 11.8 of GDP in 2016—but are expected to decline as a percent of GDP over the medium term, reaching 9.2 percent of GDP in 2021.

Evolution of Selected Macroeconomic Indicators

	2013	2014	2015	2016	Long term [1/]
Real GDP growth					
Current DSA	3.6	4.3	6.5	6.6	5.5
Previous DSA	3.6	4.7	5.1	5.9	5.8
Overall fiscal deficit (percent of GDP)					
Current DSA	5.5	5.0	4.8	4.2	3.0
Previous DSA	5.5	4.9	4.7	4.2	2.6
Current account deficit (percent of GDP)					
Current DSA	10.4	8.9	7.4	6.5	6.7
Previous DSA	10.4	8.8	8.2	8.7	7.4

[1/] Defined as the last 15 years of the projection period. For the current DSA update, the long term covers the years 2022-2036.

- **Inflation**. Inflation has increased in 2016, consistent with changes in commodity prices. The GDP deflator is projected at 1.8 percent in 2016 and is expected to remain at that level until 2021, a small upward revision compared to the previous DSA.
- **External financing mix and terms**. The DSA assumes that the financing mix will be consistent with a prudent borrowing strategy. Even though recent borrowing has seen an increase in semi-concessional and non-concessional borrowing, including on commercial terms in the WAEMU market, the authorities are engaged in pursuing a borrowing strategy that prioritizes concessional and semi-concessional over non-concessional financing to reduce borrowing costs and extend maturities. Consistent with this strategy, the DSA projects a moderate substitution between concessional and commercial borrowing—with two exceptions in 2021 and 2024 to rollover the outstanding 10-year Eurobonds issued in 2011 and 2014 with semi-concessional borrowing. With respect to the previous DSA, there has been a small revision upwards in the assumption on commercial borrowing, to reflect the recent greater willingness to contract non-concessional borrowing and the fact that traditional multilateral and bilateral concessional financing are unlikely to fund substantial investment planned under the PSE. As a result, the decline in the grant element of new disbursements is steeper and more in line with a country growing to middle-income status (Figure 1, first panel). The average maturity of new debt is close to 17.2 years, with 4.2 years of grace period (down from 19 and 4.7, respectively). Finally, the average cost of new borrowing is assumed to be 4 percent, consistent with the commitment in the MEFP.

- **Domestic borrowing.** Domestic debt is assumed to account for 30 percent of total public debt, 7.5 percent of which has maturity below one year. New short-term domestic debt is assumed to be issued at an average interest rate of 5 percent, while medium- and long-term domestic debt is assumed to carry a real interest rate of 6 percent with average maturity of 4.7 years, consistent with the current structure of domestic debt.

- **Discount rate**. The discount rate for this DSA is set at 5 percent.

EXTERNAL DSA

4. External debt indicators are below the thresholds under the baseline scenario and even under most stress tests (Figure 1). Public and publicly guaranteed (PPG) external debt is projected at 39.9 percent of GDP in 2016 and is estimated to decline to 35.1 percent of GDP in 2021 and below 30 percent in the long term. The ratios of the Present Value (PV) of PPG external debt—projected at 32.9 percent of GDP in 2016—show a declining trend under the baseline scenario and are comfortably below the thresholds, which take remittances into account, even under the most extreme scenarios. However, the PV debt-to-GDP ratio in 2018, under the worst case stress test, gets closer to the threshold, indicating that the authorities should carefully monitor the terms of new borrowing. Debt service ratios show two spikes in corresponding bullet repayment of the 2011 and 2014 Eurobonds, which are due, respectively, in 2021 and 2024. The financing plan assumed in the DSA already incorporates higher than usual semi-concessional borrowing in those years to rollover the Eurobonds. These two spikes do not—breach the thresholds under the baseline scenario, but they do under the stress scenarios when considering the ratio of debt service over revenue, as already shown in the previous DSA. In particular, the breaches are temporary and in the long term the debt service-to-revenue ratio remains below the thresholds under all bound tests. Under the historical scenario, all debt ratios are increasing in the long run, persistently breaching the respective thresholds. However, under this scenario, real GDP growth, the current account and net FDI flows are set at their historical averages, which are very different from recent economic trends and the current economic projections.

5. Notwithstanding the breaches of debt ratios under the historical scenario and of the debt service-to-revenue ratio under bound scenarios, Senegal remains at a low risk of debt distress. While the DSA shows some increasing concerns regarding debt sustainability, there are several factors which suggest a low risk of debt distress. First, the breaches in the debt service-to-revenue thresholds are very small and temporary, as they are due exclusively to the bullet repayment of the Eurobonds. The rollover is assumed to be financed through semi-concessional borrowing and the re-financing plan does not point out any additional vulnerability, given that debt service ratios remain under the respective threshold in the long term. Second, relative to the previous DSA, this analysis assumes a more realistic borrowing plan—with a steeper decline in the grant element of new financing—and this assumption does not translate into an adverse debt dynamics or an additional breach of debt thresholds. Third, the temporary breaches are observed under a currency depreciation scenario which may overstate the risk of debt distress in Senegal where external debt is denominated mostly in euro (45 percent of external public debt in 2015)—the pegged currency—and only 28 percent in U.S. dollars. However, the depreciation of the CFA franc remains a factor to be carefully taken into account going forward for debt management purposes. In addition, the

breaches under the historical scenario should warn about the risks of deviating from the current path of structural reforms and economic transformation envisaged under the PSE. Nonetheless, assuming that authorities will continue their reform effort, the historical scenario cannot be taken as a good benchmark to assess debt sustainability. Senegal, in fact, is already experiencing a growth trajectory which is significantly different from the recent history—real GDP is projected to average over 6.5 percent in 2015-16, compared to the historical average of 3.9. Moreover, the current account substantially improved, declining from a deficit of 10.8 percent in 2012 to 6.5 percent in 2016, and FDI inflows are expected to increase in the coming years, thanks to reforms to improve the business climate and the implementation of the Special Economic Zone (SEZ) with strong governance rules. To sustain growth in the long term it will be important to strengthen reforms and good governance.[2]

PUBLIC DSA

6. Indicators of overall public debt and debt service do not point to significant vulnerabilities stemming from domestic debt. Under the baseline scenario, the PV of total public debt increases from 50 percent in 2015 to 52.1 percent in 2016, but then is projected to moderately decline to 45.3 percent of GDP in 2021 (Figure 2 and Table 3). Over time, the ratio is projected to further decrease, reaching 41.5 percent in 2036. Overall, these ratios are higher than the ones estimated in the previous DSA, reflecting higher initial public debt and differences in financing assumptions and in the macroeconomic framework (notably slightly higher fiscal deficit and lower growth). Under the most extreme scenario of a 30 percent depreciation of the currency in 2016, debt ratios increase more than under the other scenarios in the short term, but in the long term the evolution of total public debt is similar to what is projected under the baseline scenario. On the other hand, the public debt outlook deteriorated slightly—but is still well below the public debt benchmark of 74 percent of GDP for strong performers—in the absence of further fiscal consolidation. The debt outlook when keeping real GDP growth and the primary deficit constant at their historical levels highlights the importance of fiscal consolidation and structural reforms to support strong growth and preserve debt sustainability. Under the historical scenario, the PV of total public debt is on a growing path and in 2025 is projected to be above the benchmark of 74 percent of GDP. As noted in the external DSA section, in the case of Senegal, historical averages are very different from the current baseline scenario, and they may be too pessimistic, given Senegal's improved economic performance. Overall, risks to public debt sustainability remain low, but stress tests underline the importance of making continuous efforts to reduce the fiscal deficit, strengthen economic growth and strictly control the volumes and terms of non-concessional borrowing.

[2] For a more detailed discussion, see the Staff Report and the Selected Issues paper entitled "Enabling Private Sector-Led Growth."

7. The evolution of debt service is more volatile and debt service is above 30 percent of revenues in the medium term. The public DSA exposes vulnerabilities related to debt service which are not clearly visible in the external DSA. In 2016, debt service absorbs 25.4 percent of government revenues, but this ratio is projected to be above 35 percent in 2017 and 2018. In addition, there are two other spikes in the debt service-to-revenue ratio in 2021 (39.4 percent) and 2024 (34 percent) when the two Eurobonds are due. More generally, average debt service is projected to be above 30 percent of revenue over 2016-36 under the baseline scenario and at 40 percent under the most extreme shock scenario. As in the external DSA, this illustrates the importance of implementing the planned fiscal consolidation. In addition, the authorities should continue the work done so far to extend the average maturities on domestic debt.

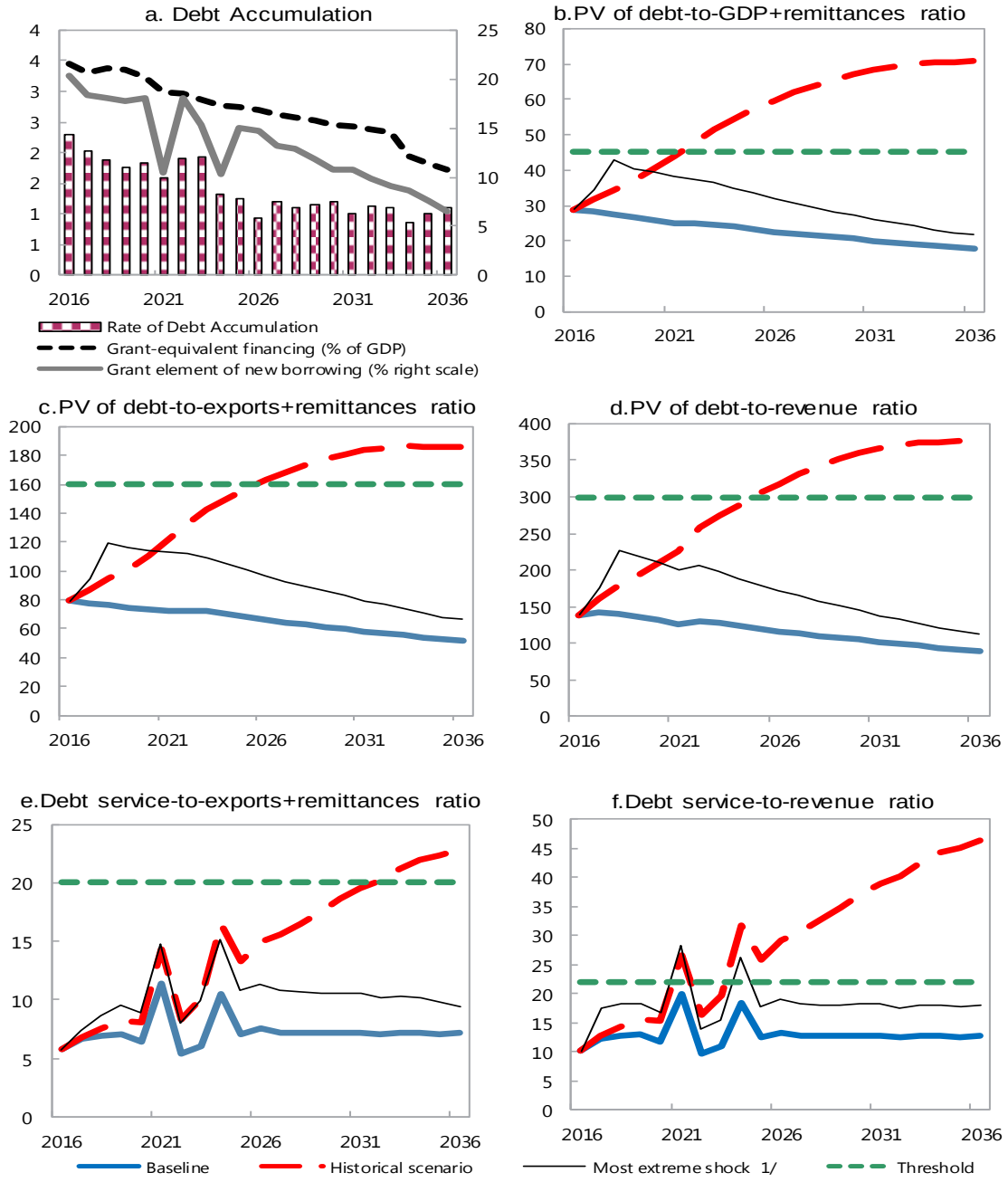
CONCLUSION

8. According to staff's assessment, Senegal continues to face a low risk of debt distress. The minor and temporary breaches of the debt service-to-revenue thresholds signal a liquidity, rather than a solvency, problem, and overall debt dynamics do not raise concerns under both the baseline and stress scenarios. However, the bound tests conducted under the external and public debt sustainability analysis—and especially the evolution of debt ratios under the historical scenario—indicate that debt sustainability hinges on continuing fiscal consolidation and on steadfastly implementing reforms to achieve high and sustained growth, as envisaged in the PSE.

9. Relative to the previous DSA, the current projections highlight that rising level of debt and debt service require a cautious approach to semi-concessional and commercial borrowing. Debt service on public debt is already high and is expected to increase in the near term, and a further deterioration of borrowing terms could put debt sustainability at risk. In this context, staff recommends a careful and continuous monitoring of financing needs and of borrowing plans, the development of a transparent pipeline of bankable projects, and a strengthening of debt management.

10. The authorities agree with the analysis in this DSA. The conclusions of the DSA were shared with the authorities who broadly concurred with the assessment and with maintaining a "low" debt risk rating. They agreed with staff that Senegal needs to reinforce its debt management capacity, especially in view of the country's gradual transition to market sources. They also expressed a strong commitment to limit non-concessional borrowing and postpone projects for which there is not adequate concessional or semi-concessional financing available.

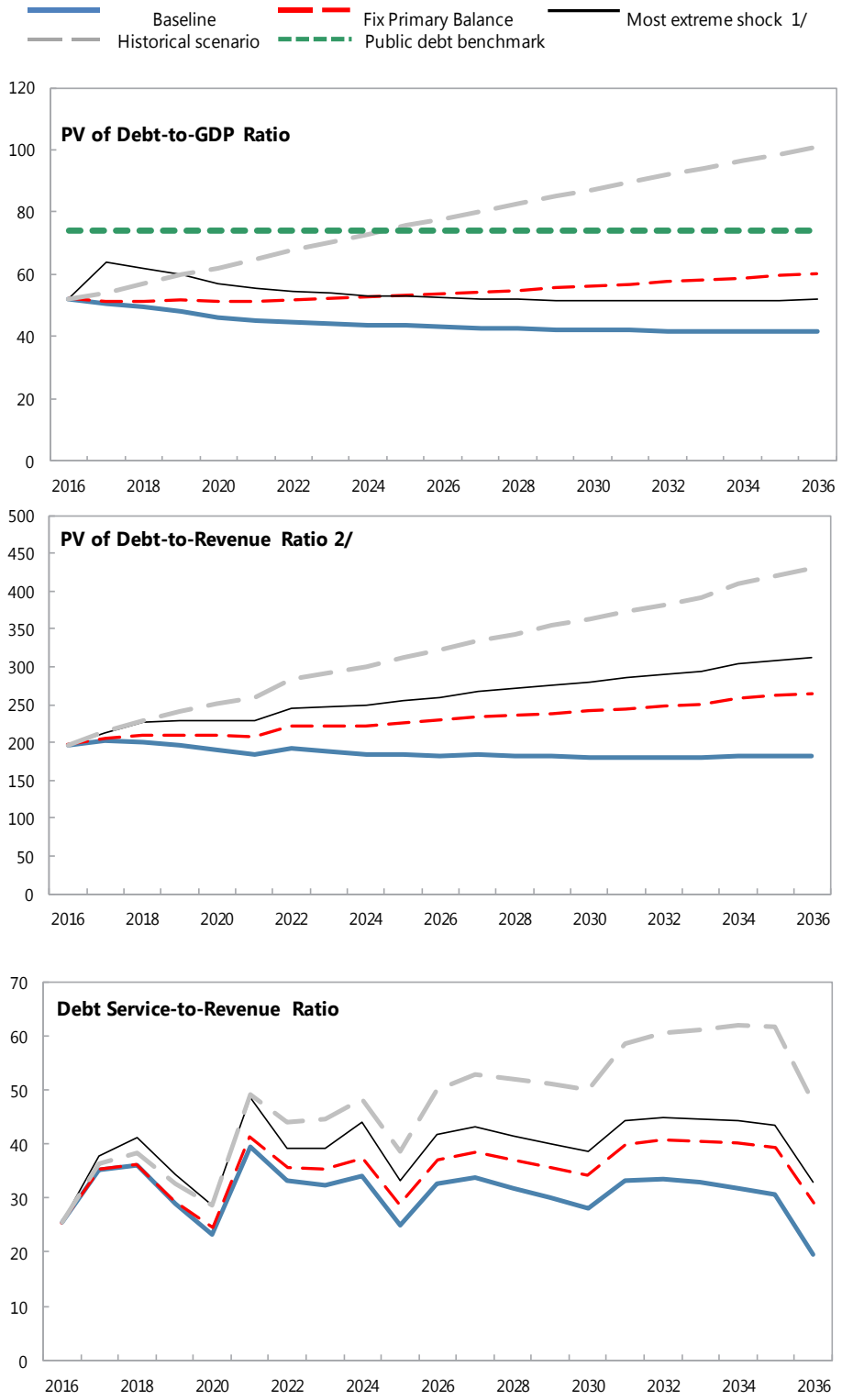
Figure 1. Senegal: Indicators of Public- and Publicly-Guaranteed External Debt Under Alternatives Scenarios, 2016-36[1]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Senegal: Indicators of Public Debt Under Alternative Scenario, 2016-36[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.
2/ Revenues are defined inclusive of grants.

Table 1. Senegal: External Sustainability Framework, Baseline Scenario, 2013-36[1/]
(Percent of GDP, unless otherwise indicated)

	Actual			Historical[6/] Average	Standard[6/] Deviation	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	**69.0**	**71.2**	**73.8**			**71.3**	**68.2**	**65.9**	**63.1**	**61.9**	**59.8**		**48.7**	**33.6**	
of which: public and publicly guaranteed (PPG)	32.6	40.3	41.1			39.9	38.7	37.5	36.5	35.8	35.1		30.4	22.6	
Change in external debt	7.9	2.2	2.5			-2.5	-3.2	-2.3	-2.8	-1.2	-2.2		-2.5	-0.8	
Identified net debt-creating flows	6.0	4.7	14.0			-0.4	0.0	-0.1	-0.1	-0.6	-1.2		0.0	1.2	
Non-interest current account deficit	**9.5**	**7.8**	**5.3**	**8.1**	**3.0**	**4.9**	**5.3**	**5.4**	**5.6**	**5.4**	**5.4**		**5.4**	**5.8**	5.5
Deficit in balance of goods and services	20.8	19.2	16.8			15.7	15.4	15.3	15.2	14.4	13.9		12.9	11.8	
Exports	28.3	28.3	29.1			27.7	27.7	27.5	27.6	27.7	27.4		27.9	30.1	
Imports	49.1	47.5	45.9			43.4	43.1	42.8	42.8	42.1	41.3		40.8	41.9	
Net current transfers (negative = inflow)	-12.5	-12.8	-12.9	-12.1	0.8	-12.4	-11.8	-11.5	-11.2	-10.5	-10.1		-8.8	-6.8	-8.2
of which: official	-0.6	-0.7	-0.6			-0.6	-0.4	-0.5	-0.7	-0.5	-0.5		-0.5	-0.7	
Other current account flows (negative = net inflow)	1.2	1.4	1.4			1.6	1.7	1.6	1.6	1.6	1.6		1.3	0.8	
Net FDI (negative = inflow)	**-1.9**	**-2.0**	**-2.3**	**-2.0**	**0.2**	**-2.4**	**-2.5**	**-2.7**	**-3.0**	**-3.5**	**-3.9**		**-3.9**	**-3.9**	-3.9
Endogenous debt dynamics 2/	**-1.6**	**-1.1**	**11.0**			**-2.9**	**-2.9**	**-2.8**	**-2.7**	**-2.6**	**-2.6**		**-1.4**	**-0.6**	
Contribution from nominal interest rate	0.9	1.2	2.2			1.6	1.6	1.6	1.5	1.5	1.4		1.2	1.0	
Contribution from real GDP growth	-2.1	-2.9	-5.2			-4.5	-4.5	-4.4	-4.2	-4.1	-4.0		-2.6	-1.7	
Contribution from price and exchange rate changes	-0.5	0.6	14.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.9**	**-2.5**	**-11.5**			**-2.1**	**-3.1**	**-2.2**	**-2.7**	**-0.5**	**-1.0**		**-2.6**	**-2.0**	
of which: exceptional financing	-0.1	1.6	0.7			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	66.8			64.3	61.5	59.4	56.4	55.1	52.7		43.1	30.1	
In percent of exports	...	...	229.4			232.5	221.6	215.6	204.6	198.5	192.3		154.2	99.8	
PV of PPG external debt	**...**	**...**	**34.1**			**32.9**	**32.0**	**31.0**	**29.8**	**28.9**	**28.0**		**24.8**	**19.1**	
In percent of exports	**...**	**...**	**117.2**			**119.0**	**115.5**	**112.5**	**108.0**	**104.4**	**102.2**		**88.8**	**63.4**	
In percent of government revenues	**...**	**...**	**153.7**			**138.9**	**143.1**	**140.9**	**136.2**	**132.4**	**126.4**		**116.7**	**89.8**	
Debt service-to-exports ratio (in percent)	**7.9**	**11.1**	**14.0**			**12.6**	**15.5**	**15.2**	**15.2**	**12.9**	**19.9**		**13.4**	**11.3**	
PPG debt service-to-exports ratio (in percent)	**6.5**	**8.0**	**9.5**			**8.7**	**10.0**	**10.2**	**10.2**	**9.2**	**16.1**		**10.1**	**8.9**	
PPG debt service-to-revenue ratio (in percent)	**9.2**	**10.6**	**12.5**			**10.1**	**12.3**	**12.8**	**12.9**	**11.7**	**19.9**		**13.3**	**12.6**	
Total gross financing need (Billions of U.S. dollars)	1.5	1.4	1.0			0.9	1.2	1.2	1.3	1.2	1.6		1.8	3.8	
Non-interest current account deficit that stabilizes debt ratio	1.6	5.6	2.8			7.3	8.4	7.7	8.4	6.6	7.5		7.9	6.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.6	4.3	6.5	3.9	1.4	6.6	6.8	7.0	7.1	7.1	7.1	6.9	5.5	5.2	5.5
GDP deflator in US dollar terms (change in percent)	0.8	-0.9	-16.5	1.2	9.9	2.1	1.9	2.2	2.4	2.6	1.8	2.1	2.0	2.4	2.2
Effective interest rate (percent) 5/	1.6	1.7	2.7	1.9	0.5	2.4	2.4	2.5	2.6	2.6	2.5	2.5	2.6	3.2	2.8
Growth of exports of G&S (US dollar terms, in percent)	6.0	3.2	-8.3	5.9	11.2	3.3	9.1	8.5	9.8	10.4	7.7	8.1	8.4	8.6	8.5
Growth of imports of G&S (US dollar terms, in percent)	5.0	-0.1	-13.9	6.9	18.7	2.8	8.1	8.6	9.6	8.0	7.0	7.3	7.8	8.1	7.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	20.4	18.3	18.0	17.8	18.1	10.5	17.2	14.8	6.4	11.6
Government revenues (excluding grants, in percent of GDP)	20.1	21.4	22.2			23.7	22.4	22.0	21.9	21.9	22.1		21.3	21.3	21.2
Aid flows (in Billions of US dollars) 7/	0.8	0.7	0.9			0.5	0.5	0.5	0.6	0.6	0.6		0.8	1.2	
of which: Grants	0.4	0.5	0.4			0.4	0.4	0.5	0.5	0.6	0.6		0.8	1.1	
of which: Concessional loans	0.4	0.2	0.5			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			3.4	3.3	3.4	3.3	3.2	3.0		2.7	1.7	2.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			55.1	52.2	53.2	53.3	53.6	40.1		52.2	38.2	47.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	14.9	15.4	13.7			14.9	16.2	17.7	19.4	21.3	23.2		34.0	71.6	
Nominal dollar GDP growth	4.4	3.4	-11.0			8.8	8.8	9.3	9.6	9.8	9.1	9.2	7.7	7.7	7.8
PV of PPG external debt (in Billions of US dollars)			4.6			4.9	5.2	5.5	5.8	6.2	6.5		8.4	13.7	
(PVt-PVt-1)/GDPt-1 (in percent)						2.3	2.0	1.9	1.8	1.8	1.6	1.9	0.9	1.1	1.2
Gross workers' remittances (Billions of US dollars)	2.1	2.2	2.0			2.1	2.2	2.3	2.4	2.5	2.6		3.3	5.1	
PV of PPG external debt (in percent of GDP + remittances)	...	...	29.8			28.9	28.3	27.4	26.5	25.9	25.2		22.6	17.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	78.2			79.1	77.9	76.7	74.6	73.4	72.5		65.9	51.2	
Debt service of PPG external debt (in percent of exports + remittance	...	...	6.3			5.8	6.7	7.0	7.1	6.5	11.4		7.5	7.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016-36
(Percent)

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to GDP ratio								
Baseline	33	32	31	30	29	28	**25**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	33	36	39	42	46	50	**68**	80
A2. New public sector loans on less favorable terms in 2016-2036 2	33	33	33	32	32	32	**31**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	33	33	34	33	31	30	**27**	21
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	33	35	40	39	37	36	**30**	20
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	33	36	39	37	36	35	**31**	24
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	33	33	33	32	31	30	**26**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	33	39	50	48	46	44	**36**	24
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	33	45	44	42	41	40	**35**	27
PV of debt-to-exports ratio								
Baseline	119	116	113	108	104	102	**89**	63
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	119	130	142	153	166	182	**242**	265
A2. New public sector loans on less favorable terms in 2016-2036 2	119	118	118	117	115	116	**112**	100
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	119	116	113	109	104	102	**89**	63
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	119	147	193	186	178	174	**141**	87
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	119	116	113	109	104	102	**89**	63
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	119	120	120	116	111	109	**93**	64
B5. Combination of B1-B4 using one-half standard deviation shocks	119	141	176	169	162	158	**127**	78
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	119	116	113	109	104	102	**89**	63
PV of debt-to-revenue ratio								
Baseline	139	143	141	136	132	126	**117**	90
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	139	161	178	193	210	226	**318**	376
A2. New public sector loans on less favorable terms in 2016-2036 2	139	147	148	147	146	143	**148**	141
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	139	149	154	149	144	138	**127**	98
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	139	158	184	178	171	163	**140**	94
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	139	160	177	171	166	158	**146**	112
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	139	148	151	146	141	135	**122**	91
B5. Combination of B1-B4 using one-half standard deviation shocks	139	175	226	219	211	201	**172**	113
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	139	203	200	194	188	179	**165**	127

Table 2. Senegal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2016-36 (concluded)
(Percent)

Debt service-to-exports ratio								
Baseline	9	10	10	10	9	16	**10**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	9	10	11	12	12	21	**22**	33
A2. New public sector loans on less favorable terms in 2016-2036 2	9	10	10	10	9	16	**10**	13
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	9	10	10	10	9	16	**10**	9
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	9	11	14	15	14	23	**17**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	9	10	10	10	9	16	**10**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	9	10	10	11	10	16	**11**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	9	11	13	14	12	21	**15**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	9	10	10	10	9	16	**10**	9
Debt service-to-revenue ratio								
Baseline	10	12	13	13	12	20	**13**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	10	13	14	16	15	27	**29**	46
A2. New public sector loans on less favorable terms in 2016-2036 2	10	12	12	13	11	19	**14**	18
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	10	13	14	14	13	22	**15**	14
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	10	12	13	15	13	21	**16**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	10	14	16	16	15	25	**17**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	10	12	13	13	12	20	**14**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	10	14	16	18	16	26	**20**	16
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	10	18	18	18	17	28	**19**	18
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	8	8	8	8	8	8	**8**	8

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline. assuming

an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Senegal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013-36
(Percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	46.9	54.4	56.9			59.1	57.2	55.9	54.8	53.3	52.4		48.6	45.1	
of which: foreign-currency denominated	32.6	40.3	41.1			39.9	38.7	37.5	36.5	35.8	35.1		30.4	22.6	
Change in public sector debt	4.1	7.4	2.6			2.2	-1.9	-1.3	-1.1	-1.6	-0.8		-0.7	-0.1	
Identified debt-creating flows	3.8	7.1	6.6			-1.0	-1.2	-1.9	-1.8	-1.6	-1.4		-0.6	-0.2	
Primary deficit	4.0	3.3	2.8	3.7	0.5	2.3	1.5	0.7	0.7	0.8	0.9	1.1	1.0	1.0	1.0
Revenue and grants	22.6	24.8	25.1			26.4	25.0	24.7	24.6	24.5	24.6		23.5	22.8	
of which: grants	2.6	3.3	2.9			2.7	2.6	2.7	2.7	2.6	2.5		2.3	1.5	
Primary (noninterest) expenditure	26.6	28.0	27.9			28.7	26.5	25.5	25.3	25.2	25.5		24.5	23.8	
Automatic debt dynamics	-0.2	3.8	3.7			-3.3	-2.7	-2.6	-2.6	-2.3	-2.3		-1.6	-1.3	
Contribution from interest rate/growth differential	1.1	0.2	-1.5			-2.4	-2.6	-2.5	-2.3	-2.2	-2.3		-1.6	-1.3	
of which: contribution from average real interest rate	2.6	2.1	1.8			1.1	1.2	1.3	1.4	1.4	1.2		1.0	1.0	
of which: contribution from real GDP growth	-1.5	-2.0	-3.3			-3.5	-3.8	-3.7	-3.7	-3.6	-3.5		-2.6	-2.2	
Contribution from real exchange rate depreciation	-1.3	3.6	5.2			-0.8	-0.1	-0.2	-0.3	-0.1	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.4	0.3	-4.0			3.2	-0.7	0.6	0.7	0.0	0.6		-0.1	0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	50.0			52.1	50.5	49.4	48.2	46.4	45.3		43.0	41.5	
of which: foreign-currency denominated	...	...	34.1			32.9	32.0	31.0	29.8	28.9	28.0		24.8	19.1	
of which: external	...	...	34.1			32.9	32.0	31.0	29.8	28.9	28.0		24.8	19.1	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	12.3	11.7	12.0			13.0	12.8	11.1	8.7	7.2	11.3		9.2	5.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	199.2			197.1	201.9	199.8	195.9	189.5	184.2		182.8	182.1	
PV of public sector debt-to-revenue ratio (in percent)	...	...	225.0			219.8	225.6	224.7	220.1	212.2	204.8		202.3	195.2	
of which: external 3/	...	...	153.7			138.9	143.1	140.9	136.2	132.4	126.4		116.7	89.8	
Debt service-to-revenue and grants ratio (in percent) 4/	17.4	20.6	27.1			25.4	35.3	36.0	28.8	23.2	39.4		32.7	19.6	
Debt service-to-revenue ratio (in percent) 4/	19.6	23.8	30.6			28.3	39.4	40.5	32.4	26.0	43.8		36.2	21.1	
Primary deficit that stabilizes the debt-to-GDP ratio	-0.1	-4.1	0.3			0.0	3.4	2.0	1.9	2.3	1.7		1.7	1.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.6	4.3	6.5	3.9	1.4	6.6	6.8	7.0	7.1	7.1	7.1	6.9	5.5	5.2	5.5
Average nominal interest rate on forex debt (in percent)	2.5	2.5	3.7	2.6	0.6	2.9	3.0	3.1	3.2	3.3	3.2	3.1	3.1	3.8	3.3
Average real interest rate on domestic debt (in percent)	9.2	7.5	3.1	4.0	4.0	4.3	4.1	4.9	5.2	5.2	4.8	4.8	4.3	3.2	4.1
Real exchange rate depreciation (in percent, + indicates depreciation	-4.3	11.1	13.4	1.2	8.8	-2.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-2.5	-1.0	0.0	1.9	3.1	1.8	1.8	1.9	1.7	1.8	1.8	1.8	2.0	2.4	2.2
Growth of real primary spending (deflated by GDP deflator, in percer	1.9	10.1	5.9	1.8	3.4	9.6	-1.3	2.7	6.6	6.6	8.2	5.4	5.5	5.7	5.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	20.4	18.3	18.0	17.8	18.1	10.5	17.2	14.8	6.4	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Senegal: Sensitivity Analysis for Key Indicators of Public Debt, 2016-36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	52	51	49	48	46	45	43	42
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	52	54	57	60	62	65	78	101
A2. Primary balance is unchanged from 2016	52	51	51	52	51	51	54	60
A3. Permanently lower GDP growth 1/	52	51	50	49	47	47	47	54
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	52	54	56	57	56	57	62	72
B2. Primary balance is at historical average minus one standard deviations in 2017-201	52	53	55	54	52	50	47	45
B3. Combination of B1-B2 using one half standard deviation shocks	52	55	58	58	57	58	61	68
B4. One-time 30 percent real depreciation in 2017	52	64	62	60	57	55	52	52
B5. 10 percent of GDP increase in other debt-creating flows in 2017	52	60	58	57	55	53	50	46
PV of Debt-to-Revenue Ratio 2/								
Baseline	197	202	200	196	189	184	183	182
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	197	215	229	241	250	259	323	429
A2. Primary balance is unchanged from 2016	197	205	208	210	209	208	229	265
A3. Permanently lower GDP growth 1/	197	203	202	199	194	190	200	237
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	197	213	226	229	229	229	260	313
B2. Primary balance is at historical average minus one standard deviations in 2017-201	197	212	223	218	211	205	201	196
B3. Combination of B1-B2 using one half standard deviation shocks	197	217	233	235	233	232	257	298
B4. One-time 30 percent real depreciation in 2017	197	256	251	243	234	225	223	228
B5. 10 percent of GDP increase in other debt-creating flows in 2017	197	239	236	231	223	216	211	204
Debt Service-to-Revenue Ratio 2/								
Baseline	25	35	36	29	23	39	33	20
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	36	38	33	29	49	50	48
A2. Primary balance is unchanged from 2016	25	35	36	29	24	41	37	29
A3. Permanently lower GDP growth 1/	25	35	36	29	24	40	35	25
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-20	25	37	39	32	27	45	41	34
B2. Primary balance is at historical average minus one standard deviations in 2017-201	25	35	37	31	26	42	35	22
B3. Combination of B1-B2 using one half standard deviation shocks	25	36	39	32	27	45	41	33
B4. One-time 30 percent real depreciation in 2017	25	38	41	34	29	49	42	33
B5. 10 percent of GDP increase in other debt-creating flows in 2017	25	35	38	33	27	44	36	22

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.